



Development and validation of a lifestyle risk index to screen for metabolic syndrome and its components in two multi-ethnic cohorts

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ABSTRACT

Background: Metabolic syndrome (MetS) is a precursor to cardiovascular diseases and type 2 diabetes. Existing MetS prediction models relied heavily on biochemical measures and those based on non-invasive predictors such as lifestyle behaviours were limited. We aim to (1) develop a weighted lifestyle risk index for MetS and (2) externally validate this index using two Asian-based cohorts in Singapore.

Methods: Using data from the Multi-Ethnic Cohort (MEC) 1 ($n = 2873$, 41% male), multiple logistic regression was used to identify predictors associated with MetS. A weighted lifestyle risk index was generated using coefficients of the selected predictors in the development cohort (MEC1). Subsequently, the performance of the lifestyle risk index in predicting the occurrence of MetS within 10 years was assessed by discrimination and calibration in an external validation cohort (MEC2) ($n = 6070$, 43% male).

Results: A lifestyle risk index for MetS with nine predictors was developed (age, sex, ethnicity, having a family history of diabetes, BMI, diet, physical activity, smoking status, and screen time). This index demonstrated acceptable discrimination in the development cohort [AUC (95% CI) = 0.74 (0.71, 0.76)] and the validation cohort [AUC (95% CI) = 0.79 (0.77, 0.81)].

Conclusion: This lifestyle risk index exhibits potential for risk stratification in population-based screening programmes. Future research could apply a similar methodology to develop disease-specific lifestyle risk indices using nationwide registry-based data.

1. Introduction

Metabolic syndrome (MetS) is defined as having three or more of these five components: abdominal obesity, increased fasting glucose, increased blood pressure, increased triglycerides, and low high-density lipoprotein (HDL) cholesterol levels (Alberti et al., 2009; Heng et al., 2006). As MetS involves a cluster of risk factors, MetS serves as an important predictor for cardiovascular diseases, stroke, and type 2 diabetes (Gami et al., 2007; Marott et al., 2016; Shin et al., 2013). For example, obesity status alone may not be sufficient as a marker of cardiometabolic health as obese individuals may be metabolically healthier and have a lower risk of cardiovascular disease than metabolically unhealthy individuals with normal weight (Iacobini et al., 2019).

The global rise in MetS cases is largely attributed to suboptimal yet modifiable lifestyle behaviours (Bassi et al., 2014; Hosseini et al., 2016).

Locally, in Singapore, the prevalence of MetS ranged from 20% to 27% (Ranasinghe et al., 2017). In comparison, the prevalence of MetS varied from 16% in Vietnam to 49% in Pakistan (Ranasinghe et al., 2017). A recent population health survey in Singapore revealed that 3% of participants consumed alcohol for >4 days a week and 10% smoked daily (Ministry of Health and Health Promotion Board, Singapore, 2022). For physical activity, only 33% of Singapore residents participated in any form of sports or exercise for at least an hour per week (Ministry of Health and Health Promotion Board, Singapore, 2022) and poor diet quality (especially low wholegrain intake) remained of concern (Health Promotion Board, Singapore, 2023). Given the increasing prevalence rates of MetS in the Asia-Pacific region, developing a MetS lifestyle risk index to identify high-risk groups with poor cardio-metabolic health among Asian adults is timely.

A recent systematic review of MetS prediction models revealed that

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many existing models relied on biochemical measures that require invasive testing (Zhang et al., 2020). Prediction models based on non-invasive predictors (e.g., lifestyle behaviours) possess great potential for individuals as risk assessment tools, for general practitioners (GPs) to provide specific lifestyle recommendations and as indicators of overall population health (Owen et al., 2010; Wang et al., 2019). Differences in the prevalence of MetS by age, sex, and ethnicity were observed, particularly among Asian adults, (Rampal et al., 2012), highlighting the need to include demographic information in MetS prediction models. In the same review on MetS prediction models, the area under the curve (AUC) of existing models ranged from 0.67 to 0.95 (Zhang et al., 2020). However, the authors concluded that it is not advisable to implement any of these models in clinical practice as they were prone to overfitting and optimism (Zhang et al., 2020). Developing a model with fewer non-invasive predictors and validating it in another population can facilitate its implementation in clinical practice.

Although it is known that health risk behaviours typically co-occur and are influenced by socioeconomic factors (Noble et al., 2015), there is limited literature on how specific combinations of lifestyle risk factors among community dwelling Asian adults affect their risk of MetS (Meader et al., 2016; Noble et al., 2015). Additionally, existing MetS models tended to assign equal weight to each lifestyle behavior but recent studies have demonstrated that these behaviors contribute differently to MetS (Yu et al., 2020; Zhang et al., 2022). Overall, developing and validating a lifestyle risk index for MetS that accounts for the relative importance of multiple health behaviors would advance this research field.

In this study, we aim to (1) develop a weighted lifestyle risk index for metabolic syndrome in a multi-ethnic Asian cohort and (2) externally validate this index in a separate Asian cohort. We applied the TRIPOD guideline (The Transparent Reporting of a multivariable prediction model for Individual Prognosis Or Diagnosis) for transparency and homogeneity in reporting our results (Collins et al., 2015).

2. Methods

This study was approved by the Institutional Review Board of National University of Singapore (NUS-IRB-2022-27) and met the institution's guidelines for protection of human subjects concerning safety and privacy. Informed consent was obtained from all participants before conducting the study. The development cohort Multi Ethnic Cohort Phase 1 study [MEC1_T1 (baseline) and MEC1_T2 (follow-up)] (Tan et al., 2018) and the validation cohort Multi-Ethnic Cohort Phase 2 follow-up (MEC2_T2) study (Whitton et al., 2021) comprise two independent cohorts of Singapore residents. Further details of the sampling approach employed and the response rates for each cohort are presented in Supplementary (Methods). Informed consent was obtained from all participants and participants with available information for both exposures and outcome were included in the analysis (Development cohort: $n = 2873$; Validation cohort: $n = 6070$) (Supplementary Figs. 1 and 2).

2.1. Predictors considered for development phase

Based on the literature on lifestyle risk indices and data availability (Barbaresko et al., 2018), we considered demographic factors (age, sex, ethnicity, highest educational attainment, and annual household income), family history of chronic diseases (heart disease, cancer and diabetes) and lifestyle behaviours (body mass index (BMI), diet (including alcohol consumption), smoking status, leisure physical activity, screen time and sitting time) as potential predictors for MetS. Further details of each predictor can be found in Supplementary Text (Methods).

2.2. Ascertainment of metabolic syndrome (MetS) status

Height, weight, waist circumference, and blood pressure were

collected for participants who attended the physical examination visit (Lim et al., 2021). Fasting blood was drawn to measure triglycerides, HDL cholesterol and glucose (Tan et al., 2018). MetS status was determined based on having three out of five components by the National Cholesterol Education Program Adult Treatment Panel III criteria and Asian recommendation for waist circumference (Alberti et al., 2009; Heng et al., 2006): (A) waist circumference > 90 cm in men and > 80 cm in women, (B) fasting triglycerides ≥ 1.7 mmol/L, (C) fasting HDL cholesterol < 1.0 mmol/L in men and < 1.3 mmol/L in women, (D) fasting glucose ≥ 6.0 mmol/L and (E) blood pressure $\geq 130/85$ mmHg. MetS status was determined prospectively for the development cohort and cross-sectionally for the validation cohort.

2.3. Statistical analyses

2.3.1. Development phase

A stepwise multivariable logistic regression (backward elimination selection approach) was used to guide the selection of predictors to be retained for development of the lifestyle risk index (Chowdhury and Turin, 2020). We considered four criteria in this selection process (in order of priority): (A) the predictor was commonly included in existing healthy lifestyle indices (Barbaresko et al., 2018), (B) the adjusted association of the predictor with MetS was in line with the literature and (C) the adjusted association of the predictor with MetS was significant and (D) several model evaluation metrics or statistical tests of (1) area under the receiver operating curve (AUC) to assess discrimination, (2) accuracy - ability to differentiate MetS cases and non-cases correctly (Baratloo et al., 2015), (3) Akaike information criterion (AIC) to compare candidate models, (4) likelihood ratio tests to assess goodness of fit between 2 nested models and (5) Hosmer-Lemeshow test for 3, 5 and 10 groups to assess goodness of fit of each candidate model.

An AUC value of 0.50 is considered as no discrimination; $0.50 < \text{AUC} < 0.70$ as poor discrimination; $0.70 \leq \text{AUC} < 0.80$ as acceptable discrimination; $0.80 \leq \text{AUC} < 0.90$ as excellent discrimination and $\text{AUC} \geq 0.90$ as outstanding discrimination (Hosmer Jr. et al., 2013). A higher accuracy value implies that a candidate model better differentiated MetS cases and non-cases (Baratloo et al., 2015). Candidate models with lower AIC values were preferred (Hosmer Jr. et al., 2013). The likelihood ratio test assessed the goodness of fit between two nested models (Lewis et al., 2011) where a p -value of < 0.05 indicates that a model with more predictors fits the data significantly better than a model with fewer predictors (Das et al., 2018). A p value of > 0.05 from the Hosmer-Lemeshow test indicates a good model fit (Hosmer Jr. et al., 2013). The results of the model evaluation metrics or statistical tests for the eight candidate models considered were shown in Supplementary Table 1.

Based on the set of predictors retained for the lifestyle risk index, the coefficients of each predictor were used to allocate points, similar to the development of Framingham risk score (Sullivan et al., 2004). Apart from age which was expressed in continuous form to facilitate the development of the lifestyle risk index, the other predictors were categorised. To ensure that there is no negative scoring for any component of the lifestyle risk index for MetS, the reference profile is a male of Malay ethnicity, without a family history of diabetes, in the highest tertile of the aHEI score (T3), never/ever smoked, lean (< 23 kg/m²), engaged in a moderate level of MVPA (0.1 to < 7.5 MET h/week) and having an average daily leisure screen time of < 2 h a day. Subsequently, the points for each predictor were summed to obtain a total score. This total score estimates one's risk of MetS in the next 10 years with a higher score representing a higher risk of MetS.

The distributions of the lifestyle risk index for MetS for each cohort and stratified by MetS status (none, 1 component met, 2 components met and MetS) were visualised using histograms. Based on their lifestyle risk index for MetS, participants were then divided into three groups of low, intermediate, and high risk/prevalence of MetS. Tertiles were chosen because the Hosmer-Lemeshow model fit showed that the final

model fitted the data well for the 3 groups ($P > 0.05$) (Supplementary Table 1). The odds of MetS were calculated for each group.

2.3.2. Validation phase

Discrimination was assessed using area under the receiver operating curves (AUC) and calibration was assessed graphically and quantitatively by Cox's slope and intercept. Further details can be found in Supplementary Text (Methods).

STATA 17 (StataCorp, 2021) software was used to develop the prediction model and calculate the lifestyle risk index. The AUC curves, calibration plots, and histograms were generated using R Studio 4.2.1.

3. Results

3.1. Characteristics of study population in the development and validation cohorts

Compared to the validation cohort, participants in the development cohort tended to be younger, with a higher proportion of participants of Malay and Indian ethnicities, currently married, without university education, less likely to be employed and without a family history of cancer ($p < 0.001$) (Table 1). For lifestyle behaviours, participants in the development cohort had a better diet quality, were leaner, spent less daily leisure screen time and engaged in lower levels of leisure time MVPA than the validation cohort. No significant differences in sex, a family history of heart disease and smoking status were observed between the two cohorts.

The incidence of MetS was 10% in the development cohort, and the prevalence of MetS was 9% in the validation cohort (Table 1). For the individual MetS components, a higher proportion of participants in the development cohort had abdominal obesity (44% in development vs. 37% in validation), low HDL cholesterol (23% vs. 20%) and high blood pressure than in the validation cohort (17% vs. 8%). No significant difference in triglyceride levels and fasting glucose were observed between the two cohorts.

3.2. Development phase: Selection of predictors for the lifestyle risk index for MetS

Age, ethnicity, having a family history of diabetes and BMI were statistically significant predictors of MetS (Table 2). There was a 2% increase in MetS odds per 1-year increase in age [OR (95% CI: 1.02 (1.01, 1.04)]. Participants of Indian ethnicity had 71% higher odds of MetS than those of Malay ethnicity [OR (95% CI: 1.71 (1.20, 2.45)]. Additionally, participants with a family history of diabetes showed 50% higher odds of MetS than those without a family history of diabetes [OR (95% CI: 1.50 (1.08, 2.08)]. Overweight participants were approximately four times more likely to develop MetS than lean participants [OR (95% CI: 4.47 (3.15, 6.34)], while obese participants were almost seven times more likely to have MetS [OR (95% CI: 7.15 (4.85, 10.55)]. The other five variables included in the lifestyle risk index for MetS (i.e., sex, diet, leisure time MVPA, smoking status, and leisure screen time) were not significantly associated with MetS. The AUC of the final model with nine predictors was 0.74 (95% CI: 0.71, 0.76) and an accuracy of 62% This indicates an acceptable discrimination of MetS cases and non-cases (Hosmer Jr. et al., 2013). Additional details about the process of selecting this model over the seven other candidate models can be found in Supplementary Text (Results).

3.3. Development phase: Lifestyle risk index for MetS

Applying the scoring metrics for each lifestyle behavior to each cohort (Table 2), the lifestyle risk index ranged from 0.5 to 18.0 for the development and validation cohorts (Fig. 1A and C). A 1-point increase in the lifestyle risk index was associated with 27% higher odds of MetS in the development cohort [OR (95% CI): 1.27 (1.22, 1.32)] and 42%

higher odds of MetS in the validation cohort [OR (95% CI): 1.42 (1.37, 1.47)] (Supplementary Table 2). Based on their lifestyle risk index for MetS, participants from each cohort were subsequently classified into tertiles of low risk (scores of 0.5–4.5), intermediate risk (scores of 5.0–10.0) and high risk (scores of 10.5–18.0) of MetS.

MetS incidence in the development cohort and MetS prevalence in the validation cohort were markedly elevated (Development: 20% in highest risk tertile vs. 3% in lowest risk tertile; Validation: 20% vs. 0.5%) (Supplementary Table 2). This is further supported by the distributions of the lifestyle risk index (stratified by MetS status) where the majority of participants with none of the MetS components had scores distributed in the lower end of the lifestyle risk index for the two cohorts (Fig. 1B and D). Conversely, the majority of participants with either 2 or 3 or more components of MetS had scores distributed in the upper end of the lifestyle risk index.

3.4. Validation phase: Discrimination and calibration of the lifestyle risk index

The AUC for the development cohort was 0.74 (95% CI: 0.71, 0.76) and 0.79 (95% CI: 0.77, 0.81) for the validation cohort (Supplementary Fig. 3). For the highest risk tertile, the sensitivity was 64% and the specificity was 71% for the development cohort; the sensitivity was 74%, and the specificity was 71% for the validation cohort.

Graphically, the calibration plot from the development cohort (Fig. 2A) was similar to that of the validation cohort (Fig. 2B). The Cox's intercept was 0.301 for the development cohort and 0.311 for the validation cohort, indicating an overall underestimation of risk/prevalence. The Cox's slope was 0.079 for the development cohort and 0.089 for the validation cohort, indicating an underestimation of low risk/prevalence and overestimation of high risk/prevalence overall. In general, there was a similar level of agreement between the observed and the predicted risk/prevalence between the two cohorts, evident by the values for Cox's slope and intercept for each cohort.

4. Discussion

In this study, we developed a weighted lifestyle risk index for MetS using nine predictors: demographic factors (age, sex, and ethnicity), family history of diabetes, and lifestyle behaviours (BMI, diet, leisure MVPA, smoking status, and leisure screen time). The lifestyle risk index demonstrated acceptable discrimination in the development and validation cohort (AUC > 0.70). A similar level of agreement between observed and predicted risk/prevalence (i.e., calibration) between the two cohorts was observed.

4.1. Development phase

While statistically significant independent predictors of MetS were age, ethnicity, having a family history of diabetes and BMI, the other lifestyle-related predictors were included to emphasize the importance of modifiable behaviours in preventing MetS (Lindström and Tuomi-lehto, 2003). This concurs with existing prediction models on MetS where age, sex, diet, leisure time physical activity, smoking status and BMI were included (Zhang et al., 2020). Alcohol consumption and socioeconomic status were not included as predictors in the existing models of MetS (Zhang et al., 2020). Alcohol use was included as part of the dietary component (aHEI-2010) of the lifestyle risk index for MetS. While links between indicators of socioeconomic status and health are well-established, these indicators are not routinely collected during healthcare visits (Chan et al., 2018). Moreover, the inclusion of socioeconomic status in a screening tool for MetS could be perceived as discriminatory in the local context. Further discussion on the other candidate models and comparisons to existing models can be found in Supplementary Text (Discussion).

The model with nine predictors was converted into a simple

Table 1

Characteristics of adult participants in the development cohort (MEC1_T1–2004 to 2010) and validation cohort (MEC2_T2–2016 to 2021) based in Singapore.

Characteristics		Development cohort MEC1_T1, n = 2873 n (%)	Validation cohort MEC2_T2, n = 6070 n (%)	P -value ¹
Age	Mean ± SD	43 ± 11	46 ± 12	6.84E-28
	Below 30	400 (14)	678 (11)	9.76E-41
	30–39	597 (21)	1330 (22)	
	40–49	1125 (39)	1770 (29)	
	50–59	583 (20)	1460 (24)	
	60 and above	168 (6)	832 (14)	
Sex	Male	1192 (41)	2611 (43)	0.173
	Female	1681 (59)	3459 (57)	
Ethnicity	Chinese	1613 (56)	4376 (72)	4.47E-51
	Malay	579 (20)	697 (12)	
	Indian	681 (24)	997 (16)	
Highest educational attainment	No formal qualifications/primary	518 (18)	867 (14)	2.06E-106
	Secondary ('O'/'N' level)	1082 (38)	1260 (21)	
	Vocational/post-secondary	835 (29)	1829 (30)	
	University and above	437 (15)	2109 (35)	
Employment in the past year	Working	2066 (72)	4733 (78)	3.19E-16
	Homemaker	574 (20)	794 (13)	
	Student/retired/unemployed	232 (8)	536 (9)	
	Never married	540 (19)	1190 (20)	3.40E-09
Marital status	Currently married	2219 (77)	4437 (73)	
	Separated/divorced/widowed	113 (4)	437 (7)	
Characteristics		Development cohort MEC1_T1, n = 2873 n (%)	Validation cohort MEC2_T2, n = 6070 n (%)	P -value ¹
Annual household income (SGD)	Less than \$2000	530 (23)	768 (14)	3.93E-81
	\$2000–\$3999	772 (34)	1195 (22)	
	\$4000–\$5999	537 (23)	1249 (23)	
	\$6000–\$9999	339 (15)	1392 (26)	
	More than \$10,000	121 (5)	835 (15)	
		127 (11)	1364 (23)	1.41E-17
With family history of cancer		399 (23)	1201 (20)	0.012
With family history of heart disease		583 (20)	- 3	- 3
With family history of diabetes ³				
Lifestyle behaviours				
Diet (aHEI-2010 score)	Mean ± SD	50.7 ± 9.4	45.8 ± 9.9	3.18E-107
	Mean (SD) of each tertile			0.999
	Tertile 1 (poorest diet quality)	40.2 ± 5.5	34.9 ± 5.2	
	Tertile 2	51.1 ± 2.3	46.0 ± 2.5	
Smoking	Tertile 3	60.8 ± 4.4	56.6 ± 4.9	
	Never smoker/ ever smoker	2499 (87)	5411 (89)	2.83E-03
	Current smoker	374 (13)	659 (11)	
BMI	Mean ± SD (kg/m ²)	23.7 ± 4.3	24.2 ± 4.5	3.43E-07
	Lean	1419 (49)	2707 (45)	1.20E-04
	Overweight	989 (34)	2295 (38)	
	Obese	465 (16)	1068 (18)	
Moderate-and-vigorous intensity leisure time physical activity	Median (p25, p75), MET-h/week	5.5 (0, 16.5)	10.0 (1.8, 23.4)	2.13E-42
	None	926 (32)	1166 (19)	1.11E-43
	0.1 to <7.5 MET-h/week	671 (23)	1480 (24)	
	≥ 7.5 MET-h/week	1276 (45)	3424 (57)	
Metabolic syndrome (MetS)		Development cohort MEC1_T1, n = 2873 n (%)	Validation cohort MEC2_T2, n = 6070 n (%)	P -value ¹
Lifestyle behaviours				
Average daily leisure screen time	Median (p25, p75)	1 (0,2)	2 (1,3)	1.32E-233
	<2 h/ day	1938 (67)	2944 (49)	2.01E-63
	≥2 h/ day	935 (33)	3126 (52)	
Components of metabolic syndrome				
Abdominal obesity	Men >90 cm, women >80 cm	1268 (44)	2255 (37)	2.74E-10
	≥ 1.7 mmol/L	452 (16)	1049 (17)	0.067
High triglycerides levels	Men <1.0 mmol/L, Women <1.3 mmol/L	667 (23)	1183 (20)	4.84E-05
	≥ 130 / 85 mmHg	488 (17)	457 (8)	4.94E-42
Low HDL cholesterol levels	≥ 6 mmol / L	185 (6)	400 (7)	0.788
High blood pressure				
Impaired fasting glucose				
Overall	None	1057 (37)	2915 (48)	2.26E-22
	1 component	909 (31)	1618 (27)	
	2 components	622 (22)	999 (17)	
	3 or more component (MetS)	285 (10)	538 (9)	

¹ Differences between the 2 cohorts using chi-square tests for categorical variables and differences in means using independent *t*-tests and differences in medians using Wilcoxon rank-sum (Mann–Whitney) test.

² Missing information for development cohort ($n = 1$ for highest education level, $n = 1$ for employment, $n = 1$ for marital status, $n = 574$ for annual household income, $n = 1756$ for family history of cancer, $n = 1124$ for family history of heart disease and $n = 1124$ for family history of diabetes) and validation cohort ($n = 6$ for highest education level, $n = 7$ for employment, $n = 6$ for marital status, $n = 631$ for annual household income, $n = 83$ for family history of heart disease and $n = 55$ for family history of cancer).

³ Family history of diabetes was not available for MEC2_T2.

Table 2

Lifestyle-based prediction model for odds of metabolic syndrome (MetS) among adult participants in the development cohort (MEC1_T1–2004 to 2010 and MEC1_T2–2011 and 2016) based in Singapore.

Intercept	Development cohort- MEC1_T1 and MEC1_T2 (n = 2873)		
	OR (95% CI)	Coefficient	Score ¹
		–4.909	
Age	1.02 (1.01, 1.04)	0.024	
Below 30			0
30–39			1.0
40–49			2.0
50–59			3.0
60 and above			4.5
Sex			
Female	1.06 (0.80, 1.41)	0.056	0.0
Male	Reference	Reference	0
Ethnicity			
Chinese	1.08 (0.75, 1.54)	0.075	0.5
Malay	Reference	Reference	0
Indian	1.71 (1.20, 2.45)	0.537	2.5
Family history of diabetes			
No	Reference	Reference	0
Yes	1.50 (1.08, 2.08)	0.404	1.5
Unknown	0.95 (0.66, 1.39)	–0.046	0.0
Diet			
Tertile 1 (poorest diet quality)	1.20 (0.85, 1.70)	0.184	1.0
Tertile 2	1.32 (0.96, 1.80)	0.276	1.0
Tertile 3	Reference	Reference	0
Smoking status			
Never smoker/ ever smoker	Reference	Reference	0
Current smoker	1.44 (0.97, 2.14)	0.364	1.5
BMI ²			
Lean (< 23 kg/m ²)	Reference	Reference	0
Overweight (≥ 23 to <27.5 kg/m ²)	4.47 (3.15, 6.34)	1.498	6.0
Obese (≥ 27.5 kg/m ²)	7.15 (4.85, 10.55)	1.967	8.0
Moderate and vigorous leisure time physical activity			
None	1.08 (0.76, 1.54)	0.079	0.5
0.1 to <7.5 MET-h/week	Reference	Reference	0
≥7.5 MET-h/week	1.06 (0.76, 1.49)	0.061	0.5
Average daily leisure screen time			
<2 h/ day	Reference	Reference	0
≥2 h/ day	0.96 (0.69, 1.34)	–0.036	0.0
Area under the curve (AUC) (95% CI)	0.74 (0.71, 0.76)		

¹ Rounded off to the nearest 0.5. A higher score indicates a higher risk of MetS.

² Based on recommended BMI cut-offs for Asian.

scoresheet using an approach similar to the development of the Framingham risk score, which is a well-established screening tool implemented in clinical practice (Sullivan et al., 2004). While individuals might already be aware of their current lifestyle behaviours, this lifestyle risk index for MetS aims to enhance their awareness. In a primary care setting, general practitioners (GPs) could utilise this index to provide personalised feedback to patients. This approach aligns with the Chronic Care Model, involving collaborative care and self-management education between the patient and the healthcare team (Bodenheimer et al., 2002). An optimum chronic care is achieved when an engaged and well-informed patient interacts with a pre-emptive healthcare team. In the local context, Healthier SG, a multi-year strategy to transform primary healthcare has been implemented recently to link up the primary healthcare system with community partners in Singapore (Foo et al., 2023). GPs enrolled in this program could develop personalised health plans with each patient using this lifestyle risk index and triage patients into different risk groups. For patients in the mid-risk to high-risk

groups, GPs can refer them to allied health professionals for specialised advice or to attend health education talks at community centres (Lim et al., 2020). For patients in the low-risk group, preventive care measures such as routine health screenings and adult vaccinations can be recommended.

This lifestyle risk index is able to distinguish between participants with and without MetS, further demonstrating its potential as a valuable, population-level screening tool. Approximately 23% of participants in the development cohort and 21% of participants in the validation cohort had 1 or 2 MetS components and scored at least 10.5 on the lifestyle risk index (high risk/prevalence group). This suggests a need to target health promotion efforts for individuals with at least one MetS component as they may be at increased risk of developing MetS subsequently. Further work (including modelling and impact studies) can be undertaken to assess its potential clinical utility and application on a population level as behavioural nudges to prevent MetS (Smith et al., 2019).

4.2. Validation phase

In the validation phase involving the discrimination and calibration of the lifestyle risk index, the AUC was within the range of AUC values (0.69 to 0.83) reported by four similar studies using the logistic regression method (Obokata et al., 2015; Pujos-Guillot et al., 2017; Yang et al., 2013; Zou et al., 2018). For the two studies with AUCs above 0.80, either components of MetS or biochemical measures were included in their models (Obokata et al., 2015; Pujos-Guillot et al., 2017). MetS components were omitted as predictors to ensure that this lifestyle risk index included only non-invasive, easily ascertained information. A lifestyle risk index should favour sensitivity over specificity as the drawback of missing a MetS case is high since appropriate lifestyle changes can be made before MetS progresses to chronic disease such as cardiovascular diseases (Dobrow et al., 2018).

While a similar level of calibration between the two cohorts was observed, there was a consistent overestimation of MetS prevalence in the validation cohort. This finding is not surprising as poor lifestyle behaviours tend to coexist in populations and are correlated with increased odds of MetS (Lee et al., 2017). Given that the Cox's intercept and slope are especially of interest for external validation (Steyerberg and Vergouwe, 2014) and values from the validation cohort were close to that of the development cohort, the lifestyle risk index is applicable in an external population.

To our knowledge, this is the first Asian multi-ethnic study to develop and validate a weighted lifestyle risk index for MetS. Other strengths of this study include utilising data from two independent cohorts representing community-dwelling Asian adults from three major ethnic groups residing in Singapore. In particular, the food frequency questionnaires (FFQs) and physical activity questionnaires were previously validated in independent samples of the local population sharing similar characteristics (Tan et al., 2018).

Several limitations are worth noting. First, the time of follow up between MEC1_T1 (where the exposures are collected) and MEC1_T2 (where the outcome MetS status was ascertained) differs for each participant [Mean (SD) of follow up: 6 (2) years]. It could be that participants with shorter follow up times may not have developed MetS unlike those who have been followed up for a longer period of time.

Second, there was a considerable non-response for the MEC1_T2 (follow up) visit, leading to a smaller sample size for the development cohort.

Third, the lifestyle risk index was developed in a prospective setting

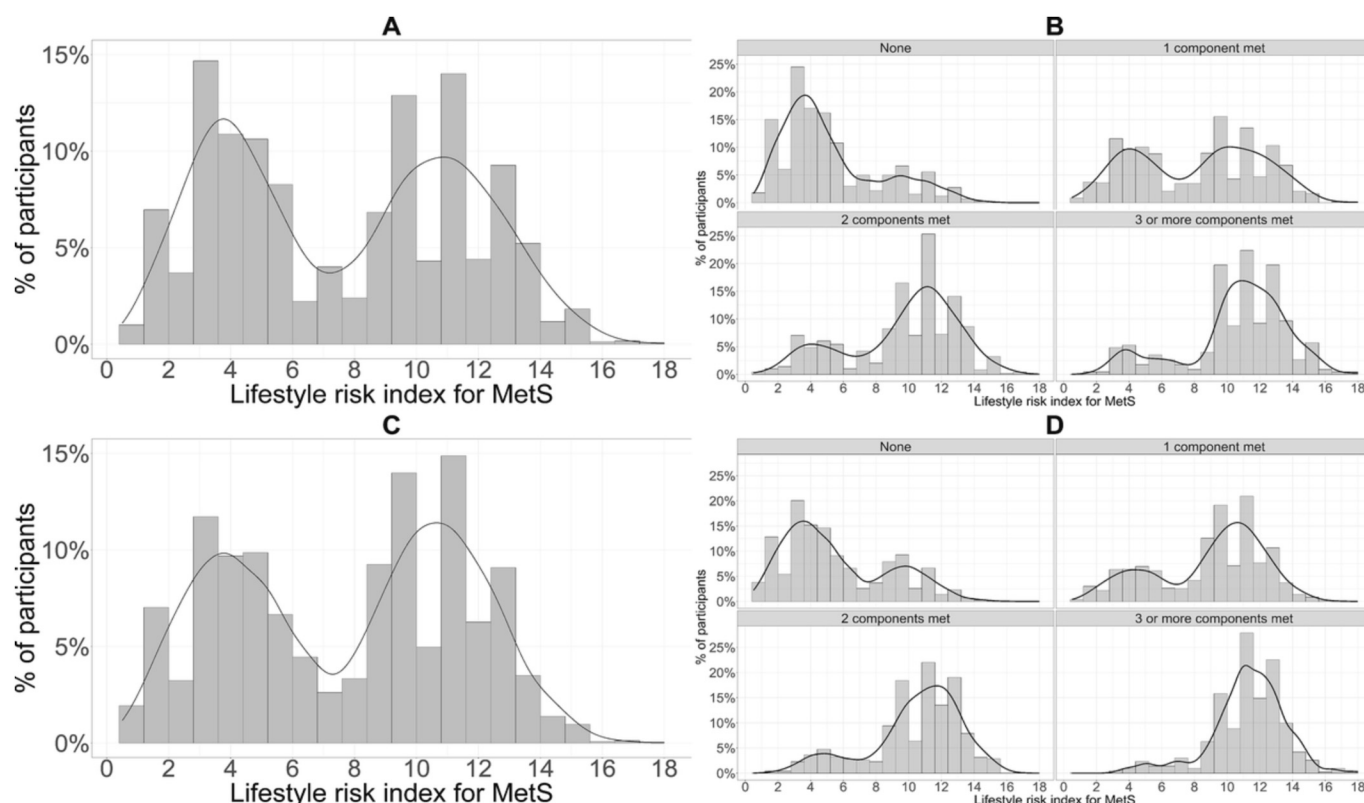


Fig. 1. Distribution of the lifestyle risk index for MetS among adult participants in the development cohort (MEC1_T1–2004 to 2010 and MEC1_T2–2011 and 2016) and validation cohort (MEC2_T2–2011 to 2016) based in Singapore. (A) Development cohort ($n = 2873$), (B) Development cohort stratified by four MetS categories (C) Validation cohort ($n = 6070$) and (D) Validation cohort stratified by four MetS categories.

and ideally, its validation should have been carried out in a similar setting. However, this lifestyle risk index was externally validated in a cross-sectional setting due to data availability at the time of study analysis. Future studies could explore developing and validating lifestyle risk indices using prospective outcomes from national disease registries with good coverage of the local population.

Fourth, there were differences in questionnaire administration between the two cohorts. Data in the development cohort was collected earlier (between 2004 and 2010) than in the validation cohort (between 2011 and 2016), and during this period, screen-based devices gained popularity. This could have led to differences in leisure screen time between the two cohorts. The applicability of the cut-off of 2 h for leisure screen time may be limited given the high leisure screen time use among local adults in the recent years (Sport Singapore, 2023).

Fifth, the diet component (calculated based on responses from the FFQs) and physical activity component (calculated based on responses from any of the 25 moderate-to-vigorous intensity physical activities) of this lifestyle index adds to respondent burden in completing the questionnaires. This may limit its utility in a clinical setting or as a self-assessment tool. Future studies may consider replacing the diet and physical activity components of the lifestyle risk index with fewer questions that are indicative of overall diet quality and physical activity levels.

In conclusion, this study developed a weighted lifestyle risk index and validated the index in an independent population. This lifestyle risk index is potentially suitable for facilitating risk communication to individuals, risk stratification in primary care and as an indicator of the overall health of the population. Future studies could explore whether the diet and physical activity components of the index could be replaced with fewer questions to enhance its utility in clinical and community settings. Additionally, a similar methodology can be applied to develop lifestyle risk indices for prospective registry-based disease outcomes.

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CRediT authorship contribution statement

Shan Xuan Lim: Data curation, Formal analysis, Project administration, Writing – original draft. **Charlie Guan Yi Lim:** Project administration, Writing – review & editing, Funding acquisition, Methodology. **Falk Müller-Riemenschneider:** Methodology, Writing – review & editing, Funding acquisition. **Rob M. van Dam:** Funding acquisition, Writing – review & editing. **Xueling Sim:** Funding acquisition, Methodology, Writing – review & editing, Project administration. **Mary Foong-Fong Chong:** Conceptualization, Funding acquisition, Supervision, Writing – review & editing. **Airu Chia:** Conceptualization, Funding acquisition, Project administration, Supervision, Writing – review & editing.

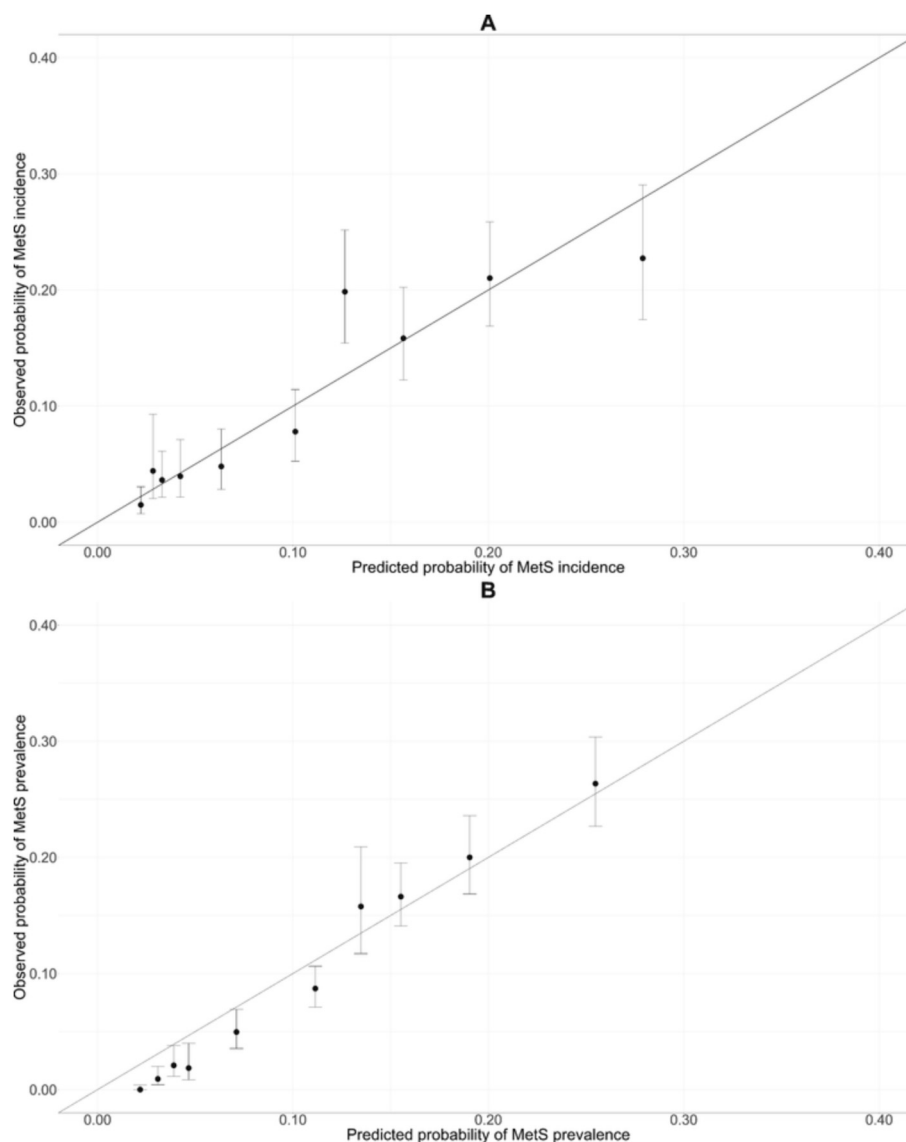


Fig. 2. Calibration plots of the lifestyle risk index for MetS among adult participants in the (A) development cohort (MEC1_T1–2004 to 2010 and MEC1_T2–2011 and 2016) (n = 2873) and (B) validation cohort (MEC2_T2–2011 to 2016) (n = 6070) based in Singapore.

Declaration of Competing Interest

The authors declare that they have no competing interests.

Data availability

Data will be made available on request.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ypmed.2023.107821>.

References

- Alberti, K.G., Eckel, R.H., Grundy, S.M., Zimmet, P.Z., Cleeman, J.I., Donato, K.A., Fruchart, J.C., James, W.P., Loria, C.M., et al., 2009. Harmonizing the metabolic syndrome: a joint interim statement of the international diabetes federation task force on epidemiology and prevention; National Heart, Lung, and Blood Institute; American Heart Association; world heart federation; international atherosclerosis society; and International Association for the Study of obesity. *Circulation* 120, 1640–1645.
- Baratloo, A., Hosseini, M., Negida, A., El Ashal, G., 2015. Part 1: simple definition and calculation of accuracy, sensitivity and specificity. *Emerg (Tehran)* 3, 48–49.
- Barbaresko, J., Rienks, J., Nöthlings, U., 2018. Lifestyle indices and cardiovascular disease risk: a Meta-analysis. *Am. J. Prev. Med.* 55, 555–564.
- Bassi, N., Karagodin, I., Wang, S., Vassallo, P., Priyanath, A., Massaro, E., Stone, N.J., 2014. Lifestyle modification for metabolic syndrome: a systematic review. *Am. J. Med.* 127 (1242.e1–10).
- Bodenheimer, T., Lorig, K., Holman, H., Grumbach, K., 2002. Patient self-management of chronic disease in primary care. *JAMA* 288, 2469–2475.
- Chan, C.Q.H., Lee, K.H., Low, L.L., 2018. A systematic review of health status, health seeking behaviour and healthcare utilisation of low socioeconomic status populations in urban Singapore. *Int. J. Equity Health* 17, 39.
- Chowdhury, M.Z.I., Turin, T.C., 2020. Variable selection strategies and its importance in clinical prediction modelling. *Fam Med Community Health* 8, e000262.
- Collins, G.S., Reitsma, J.B., Altman, D.G., Moons, K.G., 2015. Transparent reporting of a multivariable prediction model for individual prognosis or diagnosis (TRIPOD): the TRIPOD statement. *BMJ* 350, g7594.
- Das, U., Dhar, S.S., Pradhan, V., 2018. Corrected likelihood-ratio tests in logistic regression using small-sample data. *Communications in Statistics - Theory and Methods* 47, 4272–4285.
- Dobrow, M.J., Hagens, V., Chafe, R., Sullivan, T., Rabeneck, L., 2018. Consolidated principles for screening based on a systematic review and consensus process. *Cmaj* 190, E422–e29.
- Foo, C., Chia, H.X., Teo, K.W., Farwin, A., Hashim, J., Choon-Huat Koh, G., Matchar, D. B., Legido-Quigley, H., Yap, J.C.H., 2023. Healthier SG: Singapore's multi-year strategy to transform primary healthcare. *Lancet Reg Health West Pac* 37, 100861.

- Gami, A.S., Witt, B.J., Howard, D.E., Erwin, P.J., Gami, L.A., Somers, V.K., Montori, V. M., 2007. Metabolic syndrome and risk of incident cardiovascular events and death: a systematic review and meta-analysis of longitudinal studies. *J. Am. Coll. Cardiol.* 49, 403–414.
- Health Promotion Board, S, 2023. National Nutrition Health Survey 2022 Singapore. Research & Development, Singapore.
- Heng, D., Ma, S., Lee, J.J., Tai, B.C., Mak, K.H., Hughes, K., Chew, S.K., Chia, K.S., Tan, C. E., et al., 2006. Modification of the NCEP ATP III definitions of the metabolic syndrome for use in Asians identifies individuals at risk of ischemic heart disease. *Atherosclerosis* 186, 367–373.
- Hosmer Jr., D.W., Lemeshow, S., Sturdivant, R.X., 2013. *Applied Logistic Regression*, 3ed ed. John Wiley & Sons, Inc, United States of America.
- Hosseini, Z., Whiting, S.J., Vatanparast, H., 2016. Current evidence on the association of the metabolic syndrome and dietary patterns in a global perspective. *Nutr. Res. Rev.* 29, 152–162.
- Iacobini, C., Pugliese, G., Blasetti Fantauzzi, C., Federici, M., Menini, S., 2019. Metabolically healthy versus metabolically unhealthy obesity. *Metabolism* 92, 51–60.
- Lee, J.A., Cha, Y.H., Kim, S.H., Park, H.S., 2017. Impact of combined lifestyle factors on metabolic syndrome in Korean men. *J. Public Health* 39, 82–89.
- Lewis, F., Butler, A., Gilbert, L., 2011. A unified approach to model selection using the likelihood ratio test. *Methods Ecol. Evol.* 2, 155–162.
- Lim, R.B.T., Wee, W.K., For, W.C., Ananthanarayanan, J.A., Soh, Y.H., Goh, L.M.L., Tham, D.K.T., Wong, M.L., 2020. Health education and communication needs among primary care patients with prediabetes in Singapore: a mixed methods approach. *Prim. Care Diabetes* 14, 254–264.
- Lim, C.G.Y., Whitton, C., Rebello, S.A., van Dam, R.M., 2021. Diet quality and lower refined grain consumption are associated with less weight gain in a multi-ethnic Asian adult population. *J. Nutr.* 151, 2372–2382.
- Lindström, J., Tuomilehto, J., 2003. The diabetes risk score: a practical tool to predict type 2 diabetes risk. *Diabetes Care* 26, 725–731.
- Marott, S.C., Nordestgaard, B.G., Tybjaerg-Hansen, A., Benn, M., 2016. Components of the metabolic syndrome and risk of type 2 diabetes. *J. Clin. Endocrinol. Metab.* 101, 3212–3221.
- Meador, N., King, K., Moe-Byrne, T., Wright, K., Graham, H., Petticrew, M., Power, C., White, M., Sowden, A.J., 2016. A systematic review on the clustering and co-occurrence of multiple risk behaviours. *BMC Public Health* 16, 657.
- Ministry of Health and Health Promotion Board, Singapore, 2022. *Epidemiology & Disease Control Division and Policy. Research & Surveillance Group*, Singapore.
- Noble, N., Paul, C., Turon, H., Oldmeadow, C., 2015. Which modifiable health risk behaviours are related? A systematic review of the clustering of smoking, nutrition, alcohol and physical activity ('SNAP') health risk factors. *Prev. Med.* 81, 16–41.
- Obokata, M., Negishi, K., Ohya, Y., Okada, H., Imai, K., Kurabayashi, M., 2015. A risk score with additional four independent factors to predict the incidence and recovery from metabolic syndrome: development and validation in large Japanese cohorts. *PLoS One* 10, e0133884.
- Owen, N., Sparling, P.B., Healy, G.N., Dunstan, D.W., Matthews, C.E., 2010. Sedentary behavior: emerging evidence for a new health risk. *Mayo Clin. Proc.* 85, 1138–1141.
- Pujos-Guillot, E., Brandolini, M., Pétéra, M., Grissa, D., Joly, C., Lyan, B., Herquelot, É., Czernichow, S., Zins, M., et al., 2017. Systems metabolomics for prediction of metabolic syndrome. *J. Proteome Res.* 16, 2262–2272.
- Rampal, S., Mahadeva, S., Guallar, E., Bulgiba, A., Mohamed, R., Rahmat, R., Arif, M.T., Rampal, L., 2012. Ethnic differences in the prevalence of metabolic syndrome: results from a multi-ethnic population-based survey in Malaysia. *PLoS One* 7, e46365.
- Ranasinghe, P., Mathangasinghe, Y., Jayawardena, R., Hills, A.P., Misra, A., 2017. Prevalence and trends of metabolic syndrome among adults in the asia-pacific region: a systematic review. *BMC Public Health* 17, 101.
- Shin, J.A., Lee, J.H., Lim, S.Y., Ha, H.S., Kwon, H.S., Park, Y.M., Lee, W.C., Kang, M.I., Yim, H.W., et al., 2013. Metabolic syndrome as a predictor of type 2 diabetes, and its clinical interpretations and usefulness. *J. Diabetes Investig* 4, 334–343.
- Smith, T., Muller, D.C., Moons, K.G.M., Cross, A.J., Johansson, M., Ferrari, P., Fagherazzi, G., Peeters, P.H.M., Severi, G., et al., 2019. Comparison of prognostic models to predict the occurrence of colorectal cancer in asymptomatic individuals: a systematic literature review and external validation in the EPIC and UK biobank prospective cohort studies. *Gut* 68, 672–683.
- Sport Singapore, 2023. *Screen Time and Physical Activity*, Singapore.
- Steyerberg, E.W., Vergouwe, Y., 2014. Towards better clinical prediction models: seven steps for development and an ABCD for validation. *Eur. Heart J.* 35, 1925–1931.
- Sullivan, L.M., Massaro, J.M., D'Agostino Sr., R.B., 2004. Presentation of multivariate data for clinical use: the Framingham study risk score functions. *Stat. Med.* 23, 1631–1660.
- Tan, K.H.X., Tan, L.W.L., Sim, X., Tai, E.S., Lee, J.J., Chia, K.S., van Dam, R.M., 2018. Cohort profile: the Singapore multi-ethnic cohort (MEC) study. *Int. J. Epidemiol.* 47 (699–99j).
- Wang, X., Li, Y., Fan, H., 2019. The associations between screen time-based sedentary behavior and depression: a systematic review and meta-analysis. *BMC Public Health* 19, 1524.
- Whitton, C., Wong, Y.H.M., van Dam, R.M., 2021. Longitudinal associations of marital, parenting, and employment transitions with weight gain in a multi-ethnic Asian population aged 21 years and above. *Int. J. Environ. Res. Public Health* 18.
- Yang, X.H., Tao, Q.S., Sun, F., Cao, C.K., Zhan, S.Y., 2013. Setting up a risk prediction model on metabolic syndrome among 35–74 year-olds based on the Taiwan MJ health-checkup database. *Zhonghua Liu Xing Bing Xue Za Zhi* 34, 874–878.
- Yu, C.S., Lin, Y.J., Lin, C.H., Wang, S.T., Lin, S.Y., Lin, S.H., Wu, J.L., Chang, S.S., 2020. Predicting metabolic syndrome with machine learning models using a decision tree algorithm: retrospective cohort study. *JMIR Med. Inform.* 8, e17110.
- Zhang, H., Shao, J., Chen, D., Zou, P., Cui, N., Tang, L., Wang, D., Ye, Z., 2020. Reporting and methods in developing prognostic prediction models for metabolic syndrome: a systematic review and critical appraisal. *Diabetes Metab Syndr Obes* 13, 4981–4992.
- Zhang, Y., Zhang, X., Razbek, J., Li, D., Xia, W., Bao, L., Mao, H., Daken, M., Cao, M., 2022. Opening the black box: interpretable machine learning for predictor finding of metabolic syndrome. *BMC Endocr. Disord.* 22, 214.
- Zou, T.T., Zhou, Y.J., Zhou, X.D., Liu, W.Y., Van Poucke, S., Wu, W.J., Zheng, J.N., Gu, X. M., Zhang, D.C., et al., 2018. MetS risk score: a clear scoring model to predict a 3-year risk for metabolic syndrome. *Horm. Metab. Res.* 50, 683–689.